

MAJOR NATURAL GAS COMPRESSOR STATIONS

Design Basis, Design Basis Accident, and Military Action Extension

DBA-ES-GC-FC4: Natural Gas Compressor Station Facility Class

A Child Document Under
DBA-ES: Electric Sector Design Basis Framework (v0.6)

Version 0.2 — DRAFT

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For review by electric sector engineers, pipeline operators, NERC reliability analysts, FERC staff, TSA,
and policy officials responsible for electric and natural gas critical infrastructure protection.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial working draft. First publication of the DBA-ES-GC-FC4 facility-class instrument under DBA-ES sector parent (v0.6). Established facility class definition covering interstate and intrastate compressor stations in critical GenCo supply corridors. Introduced GSD (Generation Supply Duration) as the governing FC4 consequence-duration metric. Defined the gas-electric cascade as the primary consequence pathway, placing FC4 in DBA-ES rather than DBA-GAS. Instantiated all four sector-level postulated event categories with FC4-specific events. Applied DB→DBA→DBA-MA three-layer progression. Established boundary coordination with DBA-GAS-FC1 and logged the DBA-GAS §8.3 Bright Line naming correction as an open item. Incorporated Texas Winter Storm Uri (2021) and the Colonial Pipeline ransomware event (2021) as primary evidentiary anchors.
0.2	June 2026	T. Roxey	DI-3 closed. Scope boundary coordination protocol established in DBA-EN-SF v0.3 §3.3 and §3.4. §1.1, §1.6, DI-3 table row, and references section updated to reflect closure and cross-reference the governing protocol. DBA-EN-SF v0.3 §3.4 (Cross-Sector Consequence Elevation Rule) formalizes and extends the consequence-based design basis already established in FC4 §7; FC4 acceptance criteria at every facility in the class are confirmed to be subject to the elevation rule by definition, since GenCo generation curtailment carries a Tier 2 designation in the DBA-ES consequence taxonomy.

Section 1 — Purpose, Scope, and Inheritance

1.1 Purpose

This document establishes the Design Basis for major natural gas compressor stations serving critical electric generation supply corridors within the Electric Sector Design Basis Series. It is the fourth facility-class instrument in the series, following DBA-ES-HY-FC1 (large hydroelectric storage), DBA-ES-TX-FC2 (large EHV transmission substations), and DBA-ES-TX-FC3 (HVDC interconnects).

The document applies the DB→DBA→DBA-MA three-layer analytical progression to natural gas compressor stations whose primary consequence chain runs through the gas-electric cascade: loss of compressor station capacity reduces pipeline delivery pressure in a GenCo supply corridor, which curtails gas-fired generation output, which reduces bulk electric system generation capacity. This cascade pathway places FC4 instruments in DBA-ES rather than DBA-GAS, because the governing consequence is electric sector reliability, not gas infrastructure availability. The DBA-GAS series addresses the same physical facilities from the gas sector's

own design basis perspective; the two series are complementary and are coordinated at their scope boundary per DBA-EN-SF §3.3 and §3.4 (Deferred Item DI-3, now closed).

FC4 introduces Generation Supply Duration (GSD) as the governing consequence-duration metric. GSD measures the continuous period during which gas-fired generation capacity in the affected supply corridor is curtailed below its dispatchable nameplate rating due to insufficient pipeline delivery pressure. GSD is a functional metric — it measures the electric consequence of gas infrastructure failure, not the gas infrastructure failure itself, which reflects the DBA-ES series’ organizing principle of consequence-chain placement.

1.2 Facility Class Definition

A facility within the scope of DBA-ES-GC-FC4 must satisfy all three of the following criteria.

Facility type. The facility is a natural gas compressor station: a fixed installation containing one or more compressor units that increase pipeline gas pressure to maintain flow through a transmission corridor. Both gas-driven compressor units (gas turbine or reciprocating engine drivers) and electric-motor-driven compressor units are within scope. The facility may be operated by an interstate pipeline company subject to FERC jurisdiction or by an intrastate pipeline operator subject to state regulatory authority.

Corridor designation. The compressor station is located in a critical GenCo supply corridor: a pipeline segment that supplies natural gas to one or more gas-fired electric generating facilities whose combined capacity equals or exceeds 500 MW and whose loss would produce measurable bulk electric system reliability consequences in the served control area. The 500 MW threshold is calibrated to the scale at which simultaneous generation curtailment from a single supply corridor event is consequential to the electric system’s reserve margin and frequency response capability. Intrastate compressor stations satisfying this criterion are within scope, notwithstanding their state regulatory status (see Deferred Item DI-2 for regulatory coordination implications).

Single-corridor dependency. The gas-fired generating facilities in the corridor do not have an alternate gas supply path of equivalent capacity that would maintain full generation output if the FC4 compressor station were unavailable. A corridor in which the GenCo facilities have firm alternate supply from a parallel pipeline of equivalent capacity is outside the scope of this instrument; the single-corridor dependency is the vulnerability that places the compressor station within the FC4 class.

1.3 The GSD Metric

Generation Supply Duration (GSD) is introduced in this instrument as the primary consequence-duration metric for FC4. It is defined as:

GSD: The continuous period, measured from the initiating compressor station loss event, during which gas-fired generation capacity in the affected GenCo supply corridor is curtailed to zero —

below the minimum inlet pressure threshold required for any generation output — due to insufficient pipeline delivery pressure, expressed in hours or days.

GSD engages the full-curtailed condition: pipeline delivery pressure at the GenCo meter has fallen below the minimum inlet pressure required for any generation output from the affected units. Partial curtailment — in which reduced but nonzero pipeline pressure supports reduced but nonzero generation output — is not within the GSD definition at this instrument version. The partial-curtailed analysis is noted as a refinement for a future revision.

GSD does not begin at the moment of compressor station failure. The pipeline between the compressor station and the GenCo meter acts as a pressure buffer: the stored gas in the pipeline maintains delivery pressure for a period following compressor shutdown that depends on line length, diameter, and operating pressure. GSD begins when pressure at the GenCo meter falls below the minimum inlet threshold, which may be hours after the compressor station event. This buffering characteristic is analytically significant: it defines the window during which emergency response actions — compressor restart, backup supply activation, load shedding preparation — can prevent GSD from engaging at all.

GSD is the FC4 analog of TCI in FC3. Where TCI measures the duration of cross-interconnection transfer capability unavailability, GSD measures the duration of gas-fired generation capacity unavailability in the affected corridor. Both are functional metrics that measure electric consequence rather than physical infrastructure status.

1.4 Evidentiary Anchors

Two events anchor the evidentiary foundation for FC4.

Winter Storm Uri (February 2021). The February 2021 winter storm produced simultaneous failure of gas supply infrastructure and electric generation across Texas and neighboring states. Natural gas wells, gathering systems, processing facilities, and compressor stations experienced widespread freeze-related shutdowns at the moment of peak electric demand. Gas-fired power plants that depended on these supply systems for fuel lost generation capacity as pipeline pressure declined, initiating load shedding that affected approximately 4.5 million Texas customers. The event demonstrated two characteristics that are central to the FC4 design basis: the simultaneous degradation of electric generation and heating fuel supply from a single supply infrastructure failure (XC-4); and the winterization failure mode in which gas infrastructure components that were nominally operational froze during extreme cold events because they had not been designed or maintained to operate at temperatures outside their historical experience range (SE-4). Uri was not a military action, but it demonstrated with empirical precision the consequence chain that a military action against the same infrastructure could produce deliberately.

Colonial Pipeline Ransomware Attack (May 2021). The Colonial Pipeline ransomware attack caused the operator to voluntarily shut down the 5,500-mile pipeline for six days following a

cyberattack on the company’s IT systems. The pipeline itself was not physically damaged; the shutdown was a voluntary operational decision made because the operator could not safely monitor and bill for product delivery without functional IT systems. The event established a critically important principle for FC4: control system compromise — even compromise of systems not directly connected to physical pipeline operations — can produce GSD-class consequences through voluntary operational decisions rather than physical equipment failure. An adversary who understands that a pipeline operator will shut down operations when it cannot safely monitor them has an attack vector that does not require any physical action against the compressor station.

1.5 Inherited Methodology

All sector-level methodology is inherited from DBA-ES: Electric Sector Design Basis Framework (v0.6). The following table identifies inherited elements and FC4-specific adaptations.

Element	Source	FC4-Specific Adaptation
DB→DBA→DBA-MA three-layer progression	DBA-ES §3.1	Inherited without modification
Three required adaptations	DBA-ES §3.2	Inherited; consequences extend to GenCo generation curtailment beyond the compressor station boundary.
Postulated event categories: PE, SE, CE, XC	DBA-ES §3.5	FC4-specific events are instantiated in Section 3; the CE category includes pipeline SCADA and gas control systems.
Bounding event methodology	DBA-ES §4.1	The compressor unit is the governing long-lead equipment; GSD is the primary consequence-duration metric.
Natural hazard compound-event requirement	DBA-ES §4.3	Extreme cold + wind as the governing compound case, anchored on Winter Storm Uri.
Three-tier consequence taxonomy	DBA-ES §6.1	GSD defines Tier 2 duration; Tier 3 through gas-electric cascade to critical load curtailment
Long-lead-time equipment and an extended consequence regime	DBA-ES §6.2	Compressor unit lead times secondary to GSD; RT_HVLLC not the primary metric for FC4
MA extension trigger criteria	DBA-ES §1.5	FC4 satisfies Triggers 1 and 2 by definition; Trigger 5 (corridor significance) governs scope inclusion

Four MA characteristics	DBA-ES §5.2	Applied to each FC4 postulated event category in Section 5
Dual Design Basis architecture	DBA-ES §3.4	Physical DB (this document) + AI Governance DB interaction acknowledged in Section 3.6
Bright Line / Command Broker architecture	DBA-ES §3.4; DBA-DC-L1 §3.5	Bright Line is used exclusively in the ML/AI governance sense throughout this document. Note: DBA-GAS §8.3 uses “Bright Line” to mean federal coordination triggers — a naming collision to be corrected in DBA-GAS at next revision (Open Item OI-1).

1.6 Position in Hierarchy and Boundary with DBA-GAS

Document Number: DBA-ES-GC-FC4-001

Series: Electric Sector Design Basis Series — Facility Class Instruments

Parent Document: DBA-ES: Electric Sector Design Basis Framework (v0.6)

The analytical boundary between DBA-ES-GC-FC4 and DBA-GAS-FC1 (Natural Gas Transmission) is governed by consequence perspective: DBA-ES-GC-FC4 analyzes the electric sector consequences of compressor station failure, using GSD as the governing metric. DBA-GAS-FC1 analyzes the gas infrastructure consequences of the same events, using gas-sector consequence metrics. A compressor station within the FC4 scope may simultaneously be within the DBA-GAS-FC1 scope; the two instruments are analyzed separately from their respective sector perspectives. The coordination protocol for dual-scope facilities is established in DBA-EN-SF §3.3 (consequence-perspective separation rule and acceptance criteria assignment rule) and DBA-EN-SF §3.4 (cross-sector consequence elevation rule). Deferred Item DI-3 is closed by DBA-EN-SF §3.4 (v0.3).

Section 2 — Facility Profile: Natural Gas Compressor Station

2.1 Physical Configuration

A natural gas compressor station is a fixed installation containing one or more compressor units driven by gas turbines, reciprocating engines, or electric motors. The compressor units increase gas pressure from the inlet (suction) pressure to a higher outlet (discharge) pressure, maintaining the pressure gradient that drives gas flow through the pipeline. A large interstate compressor station may contain multiple units in parallel, with aggregate driver power ratings ranging from tens of thousands to hundreds of thousands of horsepower.

The compressor unit — the driver plus the compressor body — is the primary physical asset whose failure produces GSD. Gas turbine drivers are the most common large-unit configuration

on interstate pipelines; they range in capacity from approximately 5,000 to 50,000 horsepower per unit and are custom-configured for the pressure ratio and flow requirements of the specific pipeline application. Electric motor drives are common at locations with reliable utility power access and are increasingly preferred for their controllability and zero direct emissions. Reciprocating engine drives remain common at older stations and for smaller flow requirements.

The compressor station control building houses the SCADA remote terminal units, unit control panels, emergency shutdown system controllers, gas measurement equipment, and communications systems. This building is the CE vulnerability concentration point: physical damage to or cyber intrusion into the control building can disable station management without damaging the compressor units themselves.

2.2 Gas-Electric Cascade Dynamics

The gas-electric cascade that defines FC4's placement in DBA-ES operates through pipeline hydraulics. A compressor station maintains downstream delivery pressure by adding energy to the gas stream. When compression capacity is lost, the downstream pipeline segment draws on its stored gas volume — the linepack — to continue supplying demand. Linepack is finite: as demand continues to withdraw gas and no compression replenishes it, pressure declines. The rate of decline depends on the ratio of demand to stored volume, which is a function of pipeline length, diameter, and operating pressure.

For a GenCo facility at the end of a supply corridor, the minimum inlet pressure at the gas meter defines the GSD threshold. Below this pressure, the gas turbine or boiler cannot maintain sufficient fuel flow for any generation output. The time from compressor station failure to GSD onset — the linepack buffer duration — may range from a few hours (for short, high-demand corridors) to a day or more (for long, high-pressure corridors with large linepack storage). This buffer duration is the window for emergency response.

The self-reinforcing character of the cascade in gas-driven compressor stations (PE-4) is a design basis concern unique to this facility class: as pipeline pressure falls, fuel gas pressure for gas-driven unit drivers also falls, reducing compression output, which further reduces pipeline pressure. This feedback loop can accelerate pressure decay in a gas-driven corridor beyond what a simple external power loss would produce.

2.3 Regulatory Context

Interstate natural gas compressor stations are subject to FERC jurisdiction under the Natural Gas Act and to pipeline safety requirements under the Pipeline and Hazardous Materials Safety Administration (PHMSA). TSA Pipeline Security Directives SD-02C and successors establish mandatory cybersecurity requirements for critical pipeline owners and operators, including requirements for cybersecurity implementation plans, incident reporting, and vulnerability

assessments. These directives were issued following the Colonial Pipeline event and represent the current mandatory security baseline.

The TSA Security Directives do not establish a physical security design basis equivalent to NERC CIP-014. No mandatory standard requires an interstate compressor station operator to assess the physical attack scenarios addressed by this instrument or to demonstrate that the facility can withstand them. This document supplies the design basis analysis that would be needed to support such a standard.

Intrastate compressor stations are not subject to FERC jurisdiction and are regulated by state public utility commissions and state pipeline safety agencies under PHMSA oversight. The absence of a federal regulatory pathway for intrastate facilities within the FC4 scope is a structural gap identified in Deferred Item DI-2.

Following Winter Storm Uri, FERC, NERC, and PHMSA issued a joint report with recommendations for gas-electric coordination and weatherization standards. Some of these recommendations have been incorporated into NERC reliability standards and FERC orders applicable to gas-fired generators. Still, the compressor station winterization requirements remain incomplete and are not yet established as mandatory standards at the federal level.

Section 3 — Design Basis: FC4 Postulated Event Categories

3.1 Sector-Level Inheritance

The four sector-level postulated event categories — PE (Power Supply and Grid Interface), SE (Structural and Physical Integrity), CE (Control and Communications), and XC (Cross-Sector Cascade) — are inherited from DBA-ES §3.5. Each is instantiated below with FC4-specific events.

3.2 Category PE — Power Supply and Grid Interface Events

Power supply events at FC4 have a dual character that reflects the two types of compressor unit drive technology in scope. For electric-drive units, PE events are events affecting the utility power supply — directly analogous to station service loss events in other DBA-ES facility classes. For gas-driven units, PE events include loss of the fuel gas supply that powers the compressor drivers, which may be caused by conditions in the same pipeline the compressors serve. The self-reinforcing nature of gas-driven compressor fuel supply loss (PE-4) is the most analytically distinctive PE event in the DBA-ES series.

Code	Event	Initiating Condition	Governing Concern
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PE-1	Loss of station electric power — utility supply	Loss of AC power from the local electric utility to the compressor station, removing the primary supply for compressor unit drivers (electric motor drives), gas cooling equipment, control systems, and instrumentation. Gas-driven compressor units may continue operating on a pipeline gas supply if the gas pressure is maintained.	Electric-drive compressor stations are fully dependent on utility power. Loss of utility supply immediately curtails compression capacity. GSD begins when pipeline delivery pressure at the GenCo meter drops below the minimum inlet pressure threshold. Recovery requires utility power restoration or activation of backup generation.
PE-2	Loss of station electric power — extended outage	Utility power outage extending beyond the backup generation fuel endurance or UPS duration. Compressor station transitions from backup power to full shutdown. All compression capacity is lost. Pipeline pressure in the downstream supply corridor begins to decay toward ambient pressure.	Pipeline pressure decay rate governs the GSD onset and severity. Pressure decay is not instantaneous — the pipeline itself acts as a storage buffer whose capacity depends on line length, diameter, and operating pressure. GenCo generation curtailment may be delayed by hours after compressor shutdown, depending on pipeline storage capacity between the compressor station and the power plant.
PE-3	Gas supply interruption upstream of the compressor station	Loss or reduction of gas supply at the inlet to the compressor station from upstream pipeline segments, producing insufficient suction pressure for effective compression. May result from upstream compressor failures, pipeline damage, or supply source curtailment.	PE-3 is a cascade event from upstream infrastructure. The compressor station is physically undamaged but functionally unable to maintain downstream delivery pressure. GSD begins when the downstream delivery pressure falls below the GenCo minimum inlet. This event is outside the compressor station operator's control and highlights the pipeline corridor's systemic vulnerability.
PE-4	Fuel gas supply loss for gas-driven compressor units	Loss of fuel gas supply to gas-turbine or reciprocating engine compressor unit drivers. Gas-driven units draw fuel gas from the pipeline they compress; a sufficiently low inlet pressure may starve the fuel gas system and cause unit shutdown even without any external power or equipment event.	PE-4 represents a self-reinforcing failure mode: low pipeline pressure reduces fuel gas availability, which reduces compression capacity, which further reduces pipeline pressure. This positive feedback loop can accelerate pressure decay beyond what a simple compressor trip would produce.

3.3 Category SE — Structural and Physical Integrity Events

Structural and physical integrity events at FC4 concentrate on the compressor unit as the governing physical asset and on the compressor station's exposure to the extreme cold events that the Winter Storm Uri experienced, established as the governing natural hazard. SE-4 (extreme cold/winterization failure) is the natural hazard event that most directly demonstrates the gas-electric cascade: the compressor station failure mode that simultaneously curtails gas-fired generation and heating fuel supply.

Code	Event	Initiating Condition	Governing Concern
SE-1	Compressor unit failure — mechanical	Mechanical failure of a compressor unit driver (gas turbine, reciprocating engine, or electric motor) or compressor body (centrifugal or reciprocating compressor), resulting in unit trip and removal from service. A compressor station typically operates multiple units in parallel; single-unit failure reduces but does not eliminate compression capacity.	GSD onset depends on the number of units available and the required compression ratio. If the station's remaining units can maintain downstream delivery pressure above the GenCo minimum inlet threshold, GSD is not engaged. If they cannot, GSD begins immediately. The bounding SE-1 case is the simultaneous failure of all units or the failure of the single unit at a single-unit station.
SE-2	Compressor unit failure — fire or explosion	Fire or explosion in the compressor building or unit enclosure, resulting in damage to the compressor unit, associated piping, and potentially the station's control building. Compressor fires involving gas-turbine or high-pressure reciprocating units can cause severe structural damage requiring complete unit replacement.	This is the bounding SE event for FC4 from a consequence-duration perspective. Major compressor unit fire or explosion requiring unit replacement produces GSD of weeks to months, depending on unit size, manufacturer lead time, and site access conditions. Gas turbine drivers for large compressor units have lead times of six to eighteen months.
SE-3	Pipeline physical damage — rupture or leak	Physical damage to the pipeline at or adjacent to the compressor station resulting in a gas release, pressure loss, and automatic or manual isolation of the affected segment. Pipeline rupture may result from internal corrosion, external mechanical damage, material failure, or physical attack.	Pipeline isolation following a rupture removes the compressor station's function: compression into an isolated segment serves no purpose, and compression is suspended. GSD begins immediately upon isolation. Restoration requires pipeline repair, pressure testing, and regulatory reauthorization — a process that may extend from days to weeks.
SE-4	Natural hazard — extreme cold (winterization failure)	Extreme cold event producing instrument freeze-up, fuel gas line freeze, compressor unit shutdown on low-temperature protective interlocks, or	Winter Storm Uri (February 2021) is the governing evidentiary anchor for SE-4. Texas compressor stations and gas wells

		loss of control building heat resulting in instrument and control system failure. This is the Winter Storm Uri failure mode applied to compressor stations: gas-driven infrastructure that failed to deliver gas supply during the period of peak electric demand because its own supply systems were not winterized.	experienced widespread freeze-related shutdowns simultaneously with peak electric demand, reducing gas supply to gas-fired generation at the moment of greatest need. The compound event — extreme cold simultaneously increasing electric demand and reducing gas-fired generation supply — is the governing natural hazard for FC4.
SE-5	Natural hazard — compound event	Sequential or simultaneous occurrence of two natural hazard events in which recovery from the first is impaired by the second. The governing compound case for FC4 is an extreme cold event (SE-4) combined with an extreme wind event that disrupts the local utility power supply to electric-drive compressor units simultaneously. Both drivers of compression capacity are degraded at the same time.	The compound case is specifically relevant to corridor-serving compressor stations in regions where extreme cold and high wind co-occur, including the central and southern plains. The compound event eliminates both gas-driven and electric-drive compression simultaneously and may prevent personnel access for manual restoration.

3.4 Category CE — Control and Communications Events

Control and communications events at FC4 have a distinctive characteristic established by the Colonial Pipeline precedent: control system compromise can produce GSD-class consequences through voluntary operational shutdown decisions, without any physical equipment damage. This means that the CE design basis for FC4 must address not only the technical failure modes of the SCADA and control systems but also the operational decision-making framework that governs how the pipeline operator responds to control system loss.

Code	Event	Initiating Condition	Governing Concern
CE-1	Loss of pipeline SCADA at compressor station	Loss of SCADA communications between the compressor station and the pipeline operator's control center. The pipeline operator loses remote visibility into station status, compression rates, and pipeline pressure at the station. The station continues to operate under local automatic control, but the pipeline operator cannot direct compression changes, emergency shutdowns, or pressure management responses from the control center.	CE-1 is the Colonial Pipeline analog: the May 2021 Colonial Pipeline ransomware attack did not damage the pipeline physically. Still, it caused the operator to shut down operations voluntarily due to the inability to monitor and bill for product flow. Loss of SCADA at a compressor station serving a GenCo supply corridor could produce a voluntary shutdown decision with GSD consequences equivalent to physical damage.

CE-2	Cyber intrusion into compressor station control systems	Unauthorized access to compressor station control systems, including unit control panels, protective relay settings, emergency shutdown system parameters, or pipeline pressure management setpoints. Active manipulation can cause compressor unit trips, spurious emergency shutdowns, or modification of pressure management setpoints that alter downstream delivery pressure.	Compressor station control systems are increasingly networked industrial control systems with IP-layer connectivity that expands the attack surface beyond what traditional air-gapped OT environments provided. TSA Pipeline Security Directives (SD-02C and successors) address cybersecurity requirements but do not establish a design basis for the consequences of a successful intrusion.
CE-3	Loss of gas control center situational awareness	Loss of pipeline operator's gas control center visibility into the supply corridor's pressure and flow state. The gas control center cannot manage linepack, direct compression changes, or coordinate with GenCo dispatchers about anticipated supply curtailment. The GenCo operates its generators without accurate gas supply status information.	CE-3 is a gas-sector CntlCo analog: the gas control center is to the pipeline what the CntlCo is to the electric grid. Its loss under conditions of physical stress in the supply corridor is the compound event that most reliably converts a manageable physical event into an unmanaged cascade.
CE-4	GenCo — pipeline operator communications failure	Loss of direct communications between the GenCo dispatchers and the pipeline operator's gas control center. GenCo generators cannot provide scheduled gas nominations, cannot receive warning of pressure curtailments, and cannot coordinate emergency fuel switching or load reduction.	This is a governance and coordination failure as much as a technical failure. GenCo–pipeline operator coordination is governed by gas nominations and scheduling protocols that depend on functioning communications. Under military action conditions, deliberate disruption of this coordination channel amplifies the consequence of any concurrent physical event in the supply corridor.

3.5 Category XC — Cross-Sector Cascade Events

Cross-sector cascade events at FC4 operate through the gas-electric dependency that defines the facility class. XC-1 (gas-electric cascade: GenCo generation curtailment) is the primary consequence pathway and the rationale for FC4's placement in DBA-ES. XC-2 (reverse cascade) establishes the bidirectional nature of the dependency. XC-4 (heating fuel supply loss during peak demand) introduces the dual-use character of natural gas pipelines that makes winter peak demand the governing timing condition for the most severe FC4 MA scenarios.

Code	Event	Initiating Condition	Governing Concern
XC-1	Gas-electric cascade: GenCo generation curtailment	Loss of compressor station capacity reduces pipeline delivery pressure in the GenCo supply corridor below the minimum inlet pressure threshold at one or more gas-fired power plants. Power plants reduce output or trip offline as fuel supply falls below minimum operating requirements. The electric system loses generation capacity from the affected GenCo units.	This is the defining XC event for FC4 and the rationale for its placement in DBA-ES rather than DBA-GAS. GSD begins when the first GenCo unit in the corridor curtails below nameplate due to insufficient gas pressure. The cascade severity depends on the number of GenCo units in the supply corridor, their aggregate capacity, and the electric system's ability to replace the lost generation from other sources.
XC-2	Reverse cascade: electric system event reduces gas supply	An electric system event — loss of utility supply to electric-drive compressor stations, or loss of generation causing frequency decline that trips compressor station auxiliary loads — reduces compression capacity in the gas supply corridor. This initiates the gas-electric cascade (XC-1) from the electric-sector initiating event rather than the gas-sector initiating event.	XC-2 establishes the bidirectional nature of the gas-electric dependency. The cascade can run in either direction: gas supply loss reduces electric generation (XC-1), or electric supply loss reduces gas compression capacity (XC-2). Under military action conditions, an adversary can initiate the cascade from whichever end is more accessible, using the bidirectional dependency to amplify the consequence.
XC-3	Critical load curtailment through GenCo generation loss	Gas-electric cascade (XC-1) removes sufficient generation capacity from the electric system to require load shedding. Load shedding in the GenCo supply corridor's service area removes power from defense-critical facilities, hospitals, water treatment, and other critical infrastructure whose backup power posture is measured in hours.	XC-3 is the Tier 3 pathway for FC4. The cascade chain from compressor station loss to critical load curtailment may span two sector boundaries: gas sector (compressor station) → electric sector (GenCo generation loss) → critical services sector (load curtailment). The design basis must trace the full chain and establish acceptance criteria at each transition.
XC-4	Heating fuel supply loss during the peak demand period	Compressor station loss reduces natural gas pressure in the supply corridor during a winter peak demand period, curtailing gas supply not only to GenCo generation but also to residential and commercial heating customers on the same pipeline. The dual-use character of natural gas pipelines — supplying both electric generation fuel and heating fuel — means that a supply corridor disruption during winter simultaneously degrades two critical energy services.	XC-4 establishes the winter peak compound consequence that Winter Storm Uri demonstrated: loss of gas supply during extreme cold simultaneously reduces electric generation capacity and removes heating fuel from residential customers who depend on it for survival. The timing dependency is structurally similar to FC3's XC-4 (peak import dependency) — the same physical event produces qualitatively

			worse consequences at peak demand than off-peak.
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3.6 Dual Design Basis Architecture

Consistent with DBA-ES §3.4, an FC4 facility may simultaneously carry a sector-specific physical Design Basis (this document) and an AI Governance Design Basis established under the AI Governance series. The dual-DB interaction at FC4 is most significant in pipeline pressure management and compressor dispatch optimization.

Pipeline pressure management ML. Pipeline operators increasingly use ML-based pressure management systems to optimize linepack utilization and compressor dispatch across the pipeline network. These systems are ML (deterministic, testable) and operate below the Bright Line; they may operate autonomously. However, their outputs govern the linepack buffer available to GenCo facilities downstream — an ML optimization that depletes linepack in pursuit of efficiency objectives may reduce the buffer duration available to prevent GSD onset following a compressor failure. The design basis must establish minimum linepack maintenance requirements as acceptance criteria that the ML optimization system cannot override.

Operational decision support Gen AI. If Gen AI tools are used to support operational decision-making during a compressor station emergency — including decisions about whether to shut down operations following a cyber event (the Colonial Pipeline decision framework) — those tools require a human Command Broker above the Bright Line. The Command Broker must be reachable under the access-denial conditions the MA extension postulates. A voluntary shutdown decision made by a Gen AI tool without a functioning Command Broker is not an acceptable design basis response to a CE event.

Section 4 — Design Basis Accidents: Non-Adversarial Envelope

4.1 Bounding Event Methodology

The bounding event for FC4 is determined by two independent criteria: the event producing the longest GSD, and the event producing the most severe immediate impact on the electric system. The longest-GSD bounding event is SE-2 (compressor unit fire or explosion requiring unit replacement), with GSD equal to the replacement timeline of six to eighteen months for a gas turbine driver. The most severe immediate electric impact is XC-4 during winter peak demand, when simultaneous heating fuel and electric generation curtailment produces the highest consequence per unit of GSD duration.

4.2 Non-Adversarial Consequence Envelope

Tier 1 — Facility-level consequences. The bounding Tier 1 consequence is loss of all compression capacity at the facility for a duration equal to the longest unit replacement timeline (six to eighteen months for gas turbine driver replacement). Partial compression capacity may remain available if fewer than all units are affected. GSD onset is delayed by the linepack buffer duration following total compression loss.

Tier 2 — Grid-level consequences. The bounding Tier 2 consequence is full curtailment of all gas-fired generation in the affected supply corridor for the GSD duration. The electric system must replace the curtailed generation from alternative sources or implement controlled load shedding. The severity of the Tier 2 consequence depends on the proportion of electric system capacity represented by the corridor's gas-fired generation and the availability of alternative generation resources.

Tier 3 — Cross-sector consequences. The bounding Tier 3 consequence is load shedding in the service area of the curtailed generation, removing power from critical facilities whose backup power posture is measured in hours, combined with — during winter peak demand — simultaneous loss of heating fuel for residential and commercial customers. The combined consequence during the winter peak is the most severe Tier 3 case for FC4.

4.3 Extended Consequence Regime

The extended consequence regime for FC4 is defined by the intersection of GSD duration and the electric system's ability to sustain reliability obligations without the corridor's gas-fired generation. A GSD of six to eighteen months covers at least one full heating season in most US supply corridors. Resource adequacy planning must demonstrate that the electric system can sustain its reliability obligations without the corridor's generation for this duration.

The linepack buffer duration is a critical element of the extended consequence regime: it defines the window between compressor failure and GSD onset during which emergency actions can prevent escalation. The design basis must establish minimum linepack maintenance requirements — expressed as minimum pipeline pressure at the GenCo meter under defined demand conditions — as a required acceptance criterion for the FC4 facility.

Section 5 — Military Action Extension

5.1 MA Extension Applicability

All FC4 facilities satisfy at least two of the six MA extension trigger criteria established in DBA-ES §1.5 by definition. Trigger 1 fires because the governing failure mode produces Tier 2 and Tier 3 consequences. Trigger 5 (corridor significance) governs scope inclusion: the single-corridor dependency criterion in the facility class definition is itself a Trigger 5 condition. FC4 facilities serving defense-critical load areas also satisfy Trigger 4. The MA extension is mandatory for all FC4 facilities.

5.2 Application of the Four Characteristics

The four MA characteristics are applied to each postulated event category in the governing scenarios table below.

Category	Governing MA Scenario	Application of Four Characteristics	GSD Implication	Acceptance Criterion
PE	Coordinated attack on utility supply infrastructure serving electric-drive compressor stations in the corridor	Intentionality: adversary targets the utility supply infrastructure — the transmission line or substation serving the compressor station — rather than the compressor station itself, exploiting the electric-drive dependency. Persistence: an attack on the serving transmission facility may engage RT_HVLLC-class consequences at the utility level. Precision: targets the minimum utility infrastructure whose loss removes all electric-drive compression in the corridor. Denial: Area denial prevents utility repair crews from reaching the damaged transmission infrastructure.	GSD begins when electric-drive units lose power. Duration governed by utility infrastructure repair timeline — potentially days to weeks. If gas-driven backup compression is insufficient to maintain the GenCo minimum inlet pressure, GSD duration equals the utility repair timeline.	Corridor must demonstrate adequate gas-driven compression backup capacity to maintain the GenCo minimum inlet pressure for the design-basis utility outage duration.
SE	Precision kinetic attack on compressor units designed to maximize GSD through targeting the highest-lead-time replacement components	Intentionality: adversary selects gas turbine drivers as the highest-GSD elements, targeting the turbine hot section components whose replacement requires factory support and six to eighteen months lead times. Persistence: GSD equals the compressor unit replacement timeline without further adversary action. Precision: targets turbine inlet and combustion components that are most sensitive to blast and fragmentation damage. Denial: Access denial prevents damage assessment and contractor mobilization for replacement.	GSD duration equals the compressor unit replacement timeline: six to eighteen months for gas turbine driver replacement. If all units in the station are attacked, GSD equals the maximum unit replacement timeline with no partial restoration.	Physical hardening of compressor units must reduce the probability of GSD-maximizing damage from design-basis physical attack. Corridor must have identified alternative gas supply routing or backup compression that can maintain GenCo's minimum inlet pressure during GSD.
CE	Ransomware or control system intrusion causing voluntary operational shutdown — Colonial Pipeline	Intentionality: adversary targets the pipeline SCADA or the gas control center to force a voluntary shutdown decision by the pipeline operator. Persistence: GSD duration governed by forensic investigation and control system	The SCADA restoration timeline governs GSD duration. Colonial Pipeline precedent: six days of pipeline	Pipeline operator must have a defined safe operating procedure that allows continued or resumed

	paradigm applied to GenCo supply corridor	remediation timeline — potentially days to weeks. Precision: attacks the control system layer rather than the physical infrastructure, exploiting the operator’s inability to safely operate without SCADA visibility. Denial: simultaneous communications disruption prevents GenCo dispatchers from receiving warnings or coordinating emergency fuel switching.	shutdown from a ransomware attack that caused no physical damage. A more sophisticated attack targeting restoration systems simultaneously could extend GSD beyond this precedent.	compression operations under partial or complete SCADA loss, bounded by physical safety constraints. The control system restoration timeline must be within GenCo's backup fuel endurance.
XC	Attack timed to winter peak demand — Uri paradigm with adversarial intent	Intentionality: adversaries attack the winter peak demand period when heating load and electric demand are simultaneously at maximum, maximizing both the gas supply consequence and the electric generation consequence. Persistence: GSD during winter peak produces the longest period of simultaneous heating fuel and electric generation curtailment. Precision: corridor targeting selects the compressor station whose loss produces the largest combined gas-electric consequence in the supply corridor. Denial: winter weather conditions that characterize peak demand periods also impair emergency response, repair crew mobilization, and site access.	GSD consequence severity is a function of both duration and timing. A GSD of equivalent duration during winter peak produces qualitatively more severe XC-3 and XC-4 consequences than the same GSD duration at off-peak.	GenCo must demonstrate resource adequacy without corridor gas supply for the design-basis GSD duration during winter peak demand. Pipeline operator must have winterization standards sufficient to maintain compression operations through the design-basis cold weather event concurrently with a physical security event.

5.3 The Uri Paradigm with Adversarial Intent

Winter Storm Uri demonstrated the gas-electric cascade with empirical completeness. An adversary does not need to replicate a polar vortex to produce the same consequence chain; they need to replicate its effect on the compressor station infrastructure. The Uri paradigm with adversarial intent is not a theoretical construct — it is the direct application of the four MA characteristics to the failure modes that Uri demonstrated.

Intentionality replaces the random timing of a natural weather event with deliberate timing at peak demand. Persistence replaces the natural duration of a cold snap with the adversary’s choice of attack vector: a physical attack produces GSD equal to the repair or replacement timeline; a control system compromise produces GSD equal to the forensic and remediation timeline. Precision replaces the indiscriminate damage of a weather event with targeting of the specific compressor unit components — gas turbine hot section, fuel gas system — whose failure

produces the longest GSD. Denial replaces weather-induced access difficulty with deliberate area denial that prevents emergency response.

The Uri event also established the failure of coordination between gas pipeline operators and electric system operators as a contributing factor. GenCo dispatchers did not receive a timely warning of gas supply curtailment; pipeline operators did not have visibility into the electric system's dependence on their supply. An adversary can replicate this coordination failure deliberately through CE-4 (GenCo-pipeline operator communications failure), compounding the physical consequence with an information denial layer that prevents the coordinated emergency response that might otherwise arrest the cascade.

5.4 The Colonial Pipeline Paradigm

The Colonial Pipeline event established that a pipeline operator's voluntary shutdown decision — made because the operator cannot safely monitor operations without functional control systems — is a viable attack vector for producing GSD-class consequences without physical infrastructure damage. This paradigm has direct MA extension implications for FC4.

An adversary who can compromise a compressor station's SCADA system and the pipeline operator's gas control center simultaneously — CE-2 combined with CE-3 — may be able to induce a voluntary shutdown decision that produces GSD without any kinetic action. The persistence of this attack is determined by the forensic and remediation timeline, which may be days to weeks. The precision is in the targeting of the control systems rather than the physical infrastructure. The denial is in the removal of the operator's ability to assess and resume operations safely.

The design basis response to this paradigm is not primarily technical — it is procedural and architectural. The pipeline operator must have a defined safe operating procedure that allows continued or resumed compression operations under partial or complete SCADA loss, bounded by physical safety constraints. This requirement is an acceptance criterion for CE events that no current regulatory standard mandates for gas pipeline operators.

Section 6 — Consequence Analysis

6.1 Consequence Chain: SE-2 / GSD-Maximum Governing Case

Initiating event. A physical attack on the compressor station damages one or more compressor unit drivers, initiating unit shutdown. Fire suppression and emergency shutdown systems operate. If the fire involves the compressor building, control system damage may also occur.

Immediate facility consequence (Tier 1). The affected compressor units are removed from service. If all units are affected, total compression capacity is lost. Pipeline pressure in the downstream supply corridor begins to decline from the stored linepack buffer. The linepack buffer duration delays GSD onset — typically four to twelve hours for a heavily loaded corridor.

Grid consequence (Tier 2). When pipeline delivery pressure at the GenCo meter falls below the minimum inlet threshold, gas-fired generation units begin curtailing output. GSD engages. The electric system must replace curtailed generation from alternative sources. If alternative generation is insufficient, load shedding begins.

Cross-sector consequence (Tier 3). Load shedding removes power from critical facilities. During winter peak demand, simultaneous heating fuel supply curtailment for residential customers on the same pipeline begins as system pressure falls below residential delivery thresholds.

Extended consequence regime. Compressor unit replacement is initiated. GSD continues for the replacement timeline of six to eighteen months for gas turbine driver replacement. The electric system manages reliability obligations with reduced gas-fired generation capacity throughout the GSD period.

6.2 MA Extension Consequence Chain

Intentionality extends the consequence by targeting all compressor units simultaneously (eliminating partial operation fallback) and by timing the attack to winter peak demand.

Persistence is provided by the unit replacement timeline (SE attack) or control system remediation timeline (CE attack), without further adversary action at the facility.

Precision targets the gas turbine hot section components whose replacement produces the longest GSD, rather than external piping or station service infrastructure that could be repaired more quickly.

Denial prevents access for damage assessment, blocks delivery of replacement equipment, and — under the Colonial Pipeline paradigm — may extend GSD through control system compromise that prevents the operator from safely assessing whether the physical equipment is available for restart.

Section 7 — Acceptance Criteria

7.1 Tier 1 Acceptance Criteria

The compressor station must maintain a minimum linepack buffer in the downstream supply corridor sufficient to sustain GenCo's minimum inlet pressure for a minimum of twelve hours following total compression loss under peak demand conditions. This buffer duration is the minimum emergency response window required to prevent GSD onset following a compressor station event.

The facility must have defined emergency operating procedures for continued or resumed compression operations under partial or complete SCADA loss, consistent with physical safety constraints. These procedures must be tested and must be operable without communications with the gas control center.

7.2 Tier 2 Acceptance Criteria

The GenCo facilities in the supply corridor must have identified alternative gas supply arrangements — alternate pipeline sources, liquefied natural gas backup, or firm interruptible supply agreements — that can maintain minimum generation output for the design-basis GSD duration. These arrangements must be pre-contracted and physically testable, not contingent on spot market availability during an emergency.

The electric system must demonstrate resource adequacy without the corridor’s gas-fired generation for the design-basis GSD duration during winter peak demand, either through alternative generation resources or through pre-arranged demand response and emergency imports.

7.3 Tier 3 Acceptance Criteria

The maximum duration of power interruption to any defense-critical or highest-priority critical facility in the service area must not exceed that facility’s backup power posture duration. Where the most time-constrained critical facility’s backup posture is shorter than the design-basis GSD duration, the Tier 3 criterion must be satisfied through pre-arranged alternative supply, not post-event emergency response.

During winter peak demand conditions, the gas pipeline operator must have pre-arranged priority restoration protocols with the local distribution company that maintains minimum residential heating pressure in life-safety-critical service areas during a supply corridor GSD event. These protocols must be operable under the access-denial conditions the MA extension postulates.

Section 8 — Open Items and Deferred Items

Item	Description	Resolution Criterion
OI-1	DBA-GAS §8.3 Bright Line naming correction. DBA-GAS-Sector_Framework §8.3 uses the term “Bright Line” to mean federal coordination triggers between the gas sector and federal agencies — a meaning distinct from the ML/AI governance concept. This naming collision must be corrected in DBA-GAS at its next substantive revision. The federal coordination trigger concept should be renamed to a distinct term (for example, “Federal Coordination Threshold” or “Federal Notification Trigger”) and all references in DBA-GAS updated. DBA-OIL-Sector_Framework §8.3 contains the same naming collision and requires the same correction.	DBA-GAS §8.3 and DBA-OIL §8.3 revised to use a distinct term for federal coordination triggers; all cross-references updated; Bright Line used exclusively in the ML/AI governance sense across all DBA series instruments.
DI-1	GSD acceptance criterion quantitative calibration. Section 7.3 establishes GSD as the governing consequence-duration metric and identifies GenCo resource adequacy as the acceptance criterion framework. The specific GSD duration threshold above which electric generation curtailment produces unrecoverable Tier 3	GSD acceptance criterion calibrated for each FC4 corridor as part of site-specific design basis implementation, following

	consequences depends on the affected corridor’s GenCo capacity mix, the electric system’s alternative generation resources, and the critical load profile in the service area. This quantitative calibration is corridor-specific and cannot be completed at the series level.	guidance to be developed in a companion implementation guide.
DI-2	Intrastate compressor station regulatory coordination. FC4 includes intrastate compressor stations in critical GenCo supply corridors within its scope. Intrastate pipeline facilities are not subject to FERC jurisdiction and are regulated by state public utility commissions and state pipeline safety agencies. The design basis requirements established in this instrument are not directly enforceable against intrastate operators through existing federal regulatory pathways. A coordination mechanism between federal (FERC, TSA, DOE) and state authorities for FC4 intrastate facilities must be identified.	Coordination mechanism between federal and state authorities for FC4 intrastate facilities identified and established; design basis requirements adapted for state regulatory context.
DI-3	CLOSED. DBA-GAS-FC1 scope boundary coordination. The coordination protocol for dual-scope compressor station facilities has been established in DBA-EN-SF v0.3 §3.3 and §3.4. The consequence-perspective separation rule governs which instrument analyzes which consequence. The acceptance criteria assignment rule governs which instrument’s criteria are binding for each consequence tier. The cross-sector consequence elevation rule governs acceptance criteria tightening where the FC4 facility is the sole energy supply path for a Tier 2 or Tier 3 DBA-designated dependent facility. Both instruments may proceed to v0.2 development.	CLOSED. Protocol established in DBA-EN-SF v0.3 §3.3 and §3.4. Cross-reference DBA-GAS-FC1 v1.4 §1.3 (boundary acknowledgment).
DI-4	Winterization design basis standard. SE-4 identifies extreme cold as the governing natural hazard for FC4 and Winter Storm Uri as the evidentiary anchor. Specific winterization design basis standards — minimum operating temperature thresholds, heat tracing requirements, fuel gas supply protection specifications — that would have prevented the Uri failure modes are not yet established at the series level. These standards require coordination with PHMSA, FERC, and NERC, who have overlapping jurisdiction over gas-fired generation winterization following Uri.	Winterization design basis standards coordinated with PHMSA, FERC, and NERC; incorporated into FC4 instrument at next revision as a required postulated event acceptance criterion.
DI-5	Dual-DB AI Governance interaction points. Section 3.6 acknowledges the dual-DB architecture per DBA-ES §3.4. The specific AI governance interaction points for FC4 — ML systems used in pipeline pressure management, compressor dispatch optimization, and gas supply forecasting — are identified but not yet fully analyzed. The Bright Line boundary between ML (autonomous) and Gen AI (Command Broker required) applications must be drawn explicitly for FC4 pipeline control system AI contexts.	AI governance interaction points for FC4 pipeline control system AI applications identified and mapped to Bright Line / Command Broker architecture at next revision.

Section 9 — References

Regulatory and Standards References

Federal Energy Regulatory Commission, North American Electric Reliability Corporation, and Pipeline and Hazardous Materials Safety Administration. “The February 2021 Cold Weather

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North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

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Technical References

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Federal Energy Regulatory Commission. “Natural Gas Supply/Demand Balance for Winter 2021–2022.” Staff White Paper. Washington, D.C.: FERC, October 2021.

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Series Parent and Cross-Reference Documents

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Version 0.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-GAS-Sector_Framework. Version 1.4 — DRAFT. Eclectic Technologies, May 2026. (Companion gas-sector instrument; scope boundary coordination protocol established in DBA-EN-SF v0.3 §3.3 and §3.4; DI-3 closed.)

Roxey, Tim. DBA-ES-TX-FC2: Large EHV Transmission Substation. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-TX-FC3: HVDC Interconnect. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.